Design Brief â€“ Review 2

â€“8: maximum 4 A4 pages (font 11, margins 2 cm). Appendix A is not counted. Remove all grey guidance text and example rows, then export to PDF. File name: SEP490_<ProjectCode>_R2_DesignBrief.pdf

0. Header

Project code	FA26SE249
Project name	PrintGrid: Development of a Distributed 3D Printing Fulfillment and Scheduling Platform for a Network of Independent Printing Labs
Team	Nguyen Xuan Tan (SE181599, Leader) Ã‚Â· Dau Nguyen Bao Tuan (SE185029) Ã‚Â· Nguyen Quang Minh (SE181684) Ã‚Â· Truong Quang Nhat (SE181682)
Supervisor	Nguyen Tan Phuc (phucnt40@fpt.edu.vn)
Repository link (tag review-2)	https://github.com/tannguyen51/PrintGrid Ã¢â‚¬â commit ed231b7 (tag review-2 to be pushed)
Version	1.0
Date	02/11/2026

1. Scope

1.1 Registration Changes

(Record of Changes). Write "No changes" if none.

Section	Current	Proposed	Reason
3.2 b/c - Payment	Full payment before production	Deposit gates production; balance gates packing; forfeit policy	Review 1 Ã‚Â§I.4; BR-PAY-007/008
3.2 g - Team	Five members	Four members	Roster; decision 06/10

1.2 Out of Scope

Item	Reason
Carrier API integration and live tracking	Manual waybill only; no carrier contract
Resin and SLS printing	MVP restricted to FDM so slicing, estimation and QC stay verifiable
Native mobile apps, lab e-wallets, automated payouts, in-platform 3D modelling	Resource limit; payouts are computed in-system, settled by runbook

2. System at a Glance

, the users, the core value. Link to the architecture diagram in the SDD.

3. Key Design Decisions

ID	Decision	Options considered	Chosen option and rationale	Owner	Related FR / diagram
D1	Modular monolith, schema per module, no cross-module cascade	Microservices; layered monolith; modular monolith	One deployable, six modules, in-process events, extraction path kept; fits four people.	Tan	NFR-MAINT-004; Report 4 Ã‚Â§1.2
D2	Database-enforced no double booking	App lock; serializable; PostgreSQL EXCLUDE	EXCLUDE on queue key and time range; 23P01 maps to 409; works at machine or workshop level.	Minh	FR-SCHED-006; NFR-REL-004
D3	Draft quote with frozen version and staff gate	Lead-time table; auto-publish; draft plus review	Engine drafts from a frozen version and speculative placement; Ops publishes within a band; audit stored.	Tan	FR-CUST-006; FR-SCHED-005/010
D4	Lab self-QC proof gates hub dispatch	Hub-only QC; self-QC without gate; proof plus Ops approval	Completion needs actuals and photo proof; Ops approval gates batch sealing; rejection returns to InProgress.	Nhat	FR-LAB-005; FR-HUB-001/003
D5	Evidence-first file handover on object storage	File server; signed links; object storage with hash chain	MinIO plus SHA-256 at upload, receipt email, partner hash acknowledgement and access audit.	Tuan	FR-CUST-002; NFR-SEC-007

4. Top Risks

ID	Risk	Impact	How we verify or mitigate	Owner
R1	Scheduling promise fails on real lab data	Missed dates; weak research result	Simulator 10/25/50 labs with faults; calibration per machine model; partner-lab pilot; MAPE alert at 25%.	Minh
R2	No gateway; Ops records deposit and balance	No online checkout; demo diverges from FR-CUST-012	Gateway sandbox for both legs with idempotency and reconciliation; Ops bridge only as fallback.	Tan
R3	File streamed without an expiring URL	IP leak; loss of trust	Pre-signed URL for job plus 4 h; revoke on completion; audit sample; cross-lab and expiry tests.	Tuan

5. Known Gaps

Mismatches between the registration form, SRS, SDD and code. Include every High finding marked Kept or Planned in Form BM02.

ID	Gap	Location	Plan	Owner	Due
G-01	Public sample, pricing and network pages lack an approved FR and a registration item.	SRS matrix Ã‚Â§0; registration Ã‚Â§3.2c; SamplesController.cs:19	Submit FR-CUST-014..016 in BM03; if rejected remove the pages.	Tan	08/11/2026

ID	Gap	Location	Plan	Owner	Due
G-02	SampleModel and admin.sample_models are missing from the ERD and database design.	Report 3 Ã‚Â§Entities; Report 4 Ã‚Â§2.3; printgrid-schema.sql	Add the entity and relations; regenerate the ERD.	Tan	08/11/2026
G-06	Job file is streamed; no expiring pre-signed URL.	JobsController.cs:116-175; NFR-SEC-007	Issue a pre-signed URL for job plus 4 h; revoke on completion.	Tuan	15/11/2026

6. Proposed Mandatory Features

Within the scope in section 1. Acceptance criteria must be testable with a clear pass/fail result.

ID	Feature	Actor	Acceptance criteria (testable)	Test	Status
F-01	*Customer registration and authentication*	*Customer*	*Session on register, login and refresh; duplicate email 409; wrong password and unknown email 401; cross-user 404.*	*UT-AUTH-001..008; IT-AUTH-001..005*	*Done*
F-02	Model upload, evidence record and 3D preview	Customer	SHA-256 stored; receipt email <=1 min; STL/3MF preview; invalid file rejected with reason.	UT-MOD-001..004; IT-MOD-001..003	Done
F-03	Quote draft and staff publication	Customer; Ops Manager	Draft p95 <=60 s; out-of-band edit rejected; publish audit; frozen version reproduces price.	TC-F03-01..04	In progress
F-10	Administration, configuration and audit	System Administrator	Config effective <=1 min; every mutation audited with before/after; catalogue deprecates, never deletes.	TC-F10-01..03	Done

7. Non-Functional Requirements

Maximum 6 rows. Each NFR needs a metric, a threshold and a measurement method.

ID	Quality attribute	Metric	Threshold	Measurement method
NFR-PERF-001	Response time	Draft quote p95	<=60 s (preliminary <=5 s)	Timestamp 100 runs in k6
NFR-PERF-002	Response time	Reschedule wall-clock	<=30 s (mean <=15 s)	Simulator, 50 concurrent events
NFR-SCALE-001	Scalability	Network size	50 labs / 300 machines	Simulator run at 10/25/50 labs

8. Ownership Map

One row per member. Each member owns 2â€“3 parts and at least one main workflow (Appendix A1).

Member	Role	Owned parts (2â€“3)	Main workflow owned	Key decisions / PRs
Nguyen Xuan Tan (SE181599)	Team leader; PM/BA; backend	Customer; Admin/audit; integration	WF-05 Ops exception	D1, D3; PR #auth, #audit, #payment
Nguyen Quang Minh (SE181684)	Algorithm engineer	Scheduling engine; simulator; calibration	WF-02 Placement and offer	D2; PR #assignment, #simulator
Truong Quang Nhat (SE181682)	Backend/QA; Lab and Hub	Lab; Hub; QC and shipment	WF-03 Production; WF-04 Hub	D4; PR #lab, #hub, #shipment
Dau Nguyen Bao Tuan (SE185029)	Frontend engineer	Customer portal; Lab/Hub UI; design system	WF-01 Order to deposit	D5; PR #customer-portal, #upload, #checkout

Appendix A â€“ Defense 1.1 Targets

A1. Main Workflows

Minimum number = number of team members. A valid workflow involves at least 2 actors or at least 3 mandatory features, and changes the state of at least one main entity.

ID	Workflow	Actors	Trigger	Business outcome	Features	Entity states	Owner
WF-01	Customer order to quote and deposit	Customer; Ops Manager; Engine	Customer uploads and configures	Committed price and date plus paid deposit create a Confirmed order	F-01..F-04	Quote Pending->Draft->Ready->Converted; Order PaymentPending->Confirmed	Tuan
WF-02	Placement, offer and lab acceptance	Engine; Lab Manager; Ops Manager	Order Confirmed or placement requested	Capable lab accepts; job scheduled without double booking	F-06, F-07	Job Pending->Assigned->Accepted	Minh
WF-03	Production, self-QC and lab-to-hub batch	Lab Operator; Ops Manager; Hub QC	Lab accepts the job	Proof-approved jobs leave in a sealed batch	F-06, F-08	Job Accepted->InProgress->AwaitingInspection; QcProof None->Pending->Approved; Batch Sealed	Nhat
WF-04	Hub inspection, reprint and delivery	Hub QC; Hub Fulfillment; Customer	Batch received at hub	Inspected, consolidated and delivered order, or attributed reprint	F-08	Job AwaitingInspection->Completed/Failed->Reassigned; Order QualityCheck->Shipping->Delivered	Nhat
WF-05	Ops exception and date-change decision	Ops Manager; Customer; Engine	Forecast lateness or no feasible placement	Re-planned schedule or approved or expired date change	F-07, F-09	DateChangeRequest Proposed->Approved/Rejected/Expired	Tan

A2. Demo Scenarios

Per workflow: at least 1 main scenario and 1 exception scenario. Run on staging with seeded data.

ID	Workflow	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-01	*WF-01*	*Main*	Seeded customer, model and pricing version; deposit 50%	Login -> upload -> configure -> quote -> Ops publish -> deposit -> Confirmed	Quote Ready; order Confirmed; deposit once	Tuan
DS-02	*WF-01*	*Exception*	Published quote past expiry	Attempt checkout	Checkout rejected; re-quote required; no order	Tuan

ID	Workflow	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-03	WF-02	Main	Confirmed order; three capable labs	Rank labs -> accept best offer with hash acknowledgement	Job Assigned->Accepted; slot reserved; no overlap	Minh
DS-04	WF-02	Exception	Top lab declines or 2 h timeout	Decline; engine re-ranks	Lab excluded; job Reassigned; decision log records reason	Minh
DS-05	WF-03	Main	Accepted job at lab	Start -> complete with actuals and photo -> Ops approves -> seal -> hub receives	Job AwaitingInspection->InTransit->Completed; batch Received	Nhat
DS-10	WF-05	Exception	No customer answer within 48 h	Let request expire	Status Expired; escalation created; no silent change	Tan

A3. Functional Test Case Target

Source	Count	Minimum per item	Subtotal
Mandatory features	10	*3*	30
Main workflows (end-to-end)	5	*1*	5
State transitions of main entities	18	*1*	18
Agreed NFRs	6	*1*	6
Committed total			59

Automated in CI: at least 50% of the committed total.